

Examination plan: type-I conservatism of the interaction test

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1. Objective

The empirical type-I error of the biomarker-by-treatment interaction test (`bm:Db` under specifications A1 and A3, `bm:Dbc` under A2) runs systematically below the nominal 0.05 across the carryover-sensitivity factorial. The pooled mean over the 540 null cells of the production run is 0.039, and against roughly 270000 null replicates this is on the order of thirty standard errors below nominal. The deflation is therefore real and not a sampling artefact.

The manuscript (§4.6) attributes the conservatism in a single line to ‘the conservative AR(1) reference distribution at moderate sample sizes’. That attribution is plausible but unverified. The purpose of this examination is to localise the conservatism to one of four inference channels, attribute it to a mechanism, and produce an evidenced statement that either confirms the §4.6 attribution or replaces it.

A preliminary point has already been settled. At the null (`c_bm = 0`) the two DGP architectures collapse to a bit-identical data-generating process, confirmed by `check-null-architecture-equivalence.R`. The architecture axis therefore carries no information at the null, and the examination need not stratify by architecture.

2. Stage 0: scope and reference cell

The examination begins at a single null cell and expands only as the evidence requires.

- **Primary null cell.** `c_bm = 0`, exponential DGP, Hybrid design, `N = 70`, `t1half = 1.0`, `rho = 0.7`. This is the null counterpart of the manuscript reference configuration.
- **Replicates.** 5000 for the primary cell, giving a Monte Carlo standard error on the type-I estimate of approximately 0.0031 and resolving 0.039 against 0.050 at roughly 3.5 standard errors. 10000 replicates are held in reserve should the p-value ECDF tail require a cleaner estimate.
- **Reproducibility.** A fixed master seed, with per-replicate seeds derived as in the production driver. The complete per-replicate output is retained, not merely the rejection indicator.

3. Stage 1: instrument the per-replicate output

The production fitter `fit_spec()` in `simulation-core.R` returns only `estimate`, `std_error`, `p_value`, and `converged`. The decomposition that follows requires more. A standalone diagnostic driver, `diagnose-null-type1.R`, will reuse `simulation-core.R` for data generation but supply an augmented fitter that records, per replicate:

- the degrees of freedom `lme` assigns the interaction term, from `summary(fit)$tTable[, 'DF']`;
- the estimated `corCAR1` correlation parameter, the random-intercept standard deviation, and the residual standard deviation, from `VarCorr(fit)` and `fit$sigma`;
- the `non_positive_definite_rate` attribute returned by `generate_data()`, so that the positive-definiteness correction can be excluded as a contributor at negligible cost;
- the optimizer convergence messages.

The diagnostic driver does not modify the production pipeline.

4. Stage 2: decompose into the four inference channels

The interaction test is a Wald t-statistic, $T = \text{estimate} / \text{std_error}$, rejected when $|T|$ exceeds the t critical value at the assigned degrees of freedom. Conservatism must enter through one of four channels, and the channels are separable. From the primary 5000-replicate output we compute the following.

1. **Point-estimate bias.** The mean of `estimate` and a test of its equality to zero. The expected value is approximately zero; a non-zero mean would constitute a separate problem and would, if anything, mask the conservatism rather than cause it.
2. **Standard-error calibration.** The ratio of the mean model-reported `std_error` to the empirical standard deviation of `estimate` across replicates, together with the full distribution of `std_error` so that a uniform upward shift can be distinguished from a heavy right tail. A ratio above 1.05 implicates standard-error over-estimation.
3. **Statistic dispersion.** The standard deviation of T under the null, which should be approximately one under correct calibration. A value below one confirms a conservative reference.
4. **p-value uniformity.** The empirical CDF of the p-values against the diagonal, Kolmogorov-Smirnov and Anderson-Darling tests against Uniform(0, 1), and the type-I rate at alpha in {0.01, 0.05, 0.10}. Whether all three alpha levels are depressed or only the tail distinguishes a global scale error from a tail-shape error.

The deliverable of this stage is a four-row decomposition table and the p-value ECDF figure.

5. Stage 3: separate the DF channel from the SE channel

The p-values are recomputed from the same `estimate` and `std_error` against a standard-normal reference in place of the t reference.

- If the type-I rate rises to approximately 0.05 under the normal reference, the `lme` degrees-of-freedom rule is the cause.
- If the type-I rate remains at approximately 0.039, the degrees-of-freedom rule is excluded and the cause lies in the standard error or the joint estimate-and-SE calibration. The examination then proceeds to Stage 4.

A prior check short-circuits this stage. The assigned degrees of freedom are tabulated directly; with $N = 70$ across eight measurement occasions the level-one degrees of freedom are likely in the hundreds, in which case the t and normal references are numerically indistinguishable and the degrees-of-freedom channel is excluded immediately.

6. Stage 4: attribute the SE mis-calibration to a covariance model

The same saved null datasets are refitted under a ladder of correlation specifications, comparing the interaction standard error, the calibration ratio, and the type-I rate across them.

- **M0.** `lm(Sx ~ bm + t + Db + bm:Db)`, ignoring all within-subject correlation. Expected to be anti-conservative and serving as a lower anchor.
- **M1.** `lme` with a random intercept only, no `corCAR1`.
- **M2.** `gls` with a `corCAR1` residual only, no random intercept.
- **M3.** `lme` with both a random intercept and a `corCAR1` residual, the production model.

The data-generating process produces AR(1)-correlated factors plus a participant-level baseline term `BL` that acts as a random intercept. Specification M3 is therefore the well-specified member of the ladder, and M1 and M2 are each mis-specified. The reading is as follows. If M3 alone is conservative while M1 is calibrated, the near-redundant pairing of a random intercept with a `corCAR1` residual on only eight occasions is implicated. If M1, M2, and M3 are all conservative, the cause is the generic finite-sample plug-in Wald-t and is not specific to the redundancy.

7. Stage 4b: gold-standard calibration counterfactual (conditional)

This stage is undertaken only if Stages 2 to 4 point to the Wald approximation. The conservatism is then checked against a reference that does not rely on that approximation.

- A parametric-bootstrap or permutation null distribution for the interaction statistic, computed on a subsample of datasets. If those p-values are uniform while the Wald p-values are not, the Wald approximation is conclusively the cause.

- Optionally, an `lmer` fit with a Kenward-Roger small-sample correction by way of `pbkrtest`, accepting that `lmer` cannot carry the `corCAR1` residual and so tests only the random-intercept structure.

8. Stage 5: design dependence

The Stage 2 decomposition is repeated for the CO, Hybrid, and OLBDC designs at `c_bm = 0`, `N = 70`. Separately, each design's long-form design matrix is inspected: the correlation and variance inflation factor between `Db` and `t`, and between the `bm:Db` and `bm:t` columns. If the conservatism tracks the `Db-t` collinearity ranking across designs, design-induced collinearity is a contributing channel. If the type-I rate is uniformly near 0.039 irrespective of design, collinearity is excluded.

9. Stage 6: sample-size gradient

The null decomposition is run at `N` in `{35, 70, 140, 280}`. This is the decisive diagnostic. A finite-sample Wald artefact must attenuate as `N` grows, with the type-I rate approaching 0.05. If it does, the conservatism is benign and expected, and the manuscript may say so with evidence. If the type-I rate remains near 0.039 at large `N`, the cause is asymptotic, constitutes a structural mis-specification, and warrants a correction rather than a footnote.

10. Decision logic

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Stage 2  ratio > 1.05 ..... SE over-estimation
         SD(T) < 1, ECDF bowed ..... conservative reference confirmed
Stage 3  normal reference fixes it ..... DF rule is the cause
         normal reference does not ..... SE/calibration, go to Stage 4
Stage 4  M1 calibrated, M3 not ..... RI + corCAR1 redundancy
         M1, M2, M3 all conservative .... generic finite-sample Wald-t
Stage 5  tracks Db-t collinearity ..... design collinearity contributes
Stage 6  type-I -> 0.05 as N grows ..... finite-sample, benign
         type-I flat in N ..... asymptotic mis-specification
```

11. Deliverables

1. `diagnose-null-type1.R`, the standalone diagnostic driver.
2. A short results note containing the four-channel decomposition table, the p-value ECDF figure, the model-ladder table, the design and sample-size tables, and an attributed one-paragraph conclusion.
3. A concrete recommendation for §4.6 of the manuscript: either confirm the finite-sample attribution with the sample-size gradient as evidence, or replace it.

12. Cost and sequencing

All stages are computationally light. At the throughput observed in the development run, the 5000-replicate primary cell is on the order of a few minutes, and the model ladder and the sample-size gradient multiply that by small constants. The whole examination is minutes of compute, not hours.

The recommended sequencing is to run Stages 1 to 3 first, since they are quick and may already exclude the degrees-of-freedom channel and quantify the standard-error ratio, and then to decide whether Stages 4 to 6 are required. Stages 4b and 5 are conditional; Stage 6 should be run regardless, because it is the single most decisive test of whether the conservatism is the benign finite-sample behaviour the manuscript assumes.

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